

SUMMARY TABLE: MATHEMATICS GRADE 10

Sub-Strand 3.1: Trigonometry I — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 3.1: Trigonometry I
Driving Question	DRIVING QUESTION: How can the angle a sunbeam makes with the ground allow us to calculate the height of any tree, building, or hill in Kenya without measuring it directly?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Predict and Anchor: Why Does My Shadow Tell Me How Tall the Tree Is?	A student in Kisumu noticed that she could measure the angle of the sun with a clinometer and that both she and the mango tree cast shadows on the same sunny afternoon. Her shadow was 1.2 m and the tree shadow was 9.0 m, with a sun angle of approximately 53 degrees.	When you draw right-angled triangles with the same acute angle but different sizes, the ratio of opposite to hypotenuse stays the same, the ratio of adjacent to hypotenuse stays the same, and the ratio of opposite to adjacent stays the same. These constant ratios are called sine, cosine, and tangent of that angle.	The ratios are constant because all right-angled triangles with the same acute angle are similar triangles — their sides are proportional. This means the angle alone determines the ratio, so if we know the angle and one side we can always find the other side. That is how the 53-degree sun angle can reveal the mango tree height from just the shadow length.
Lesson 2 Sine of an Acute Angle: Reading the Language of Triangles	Learners watched a video showing that two right-angled triangles with the same 40° angle but different sizes produce the same value of opposite ÷ hypotenuse. They also observed that a surveyor in the Rift Valley uses this relationship to measure hill heights without direct measurement.	The sine of an acute angle ($\sin \theta$) is defined as opposite ÷ hypotenuse in a right-angled triangle and is a fixed value that depends only on the angle — not on the size of the triangle. For the mango tree scenario, $\sin 53^\circ \approx 0.7986$, a value that can be read directly from mathematical tables.	Learners explained that because triangles with the same angle are similar (same shape, scaled size), their side ratios are always equal — which is why $\sin 53^\circ$ is the same whether the triangle is drawn on paper or formed by a sun ray and a mango tree in Kisumu. This means knowing just the angle and one side length is enough to find any other side — bringing us one step closer to calculating the mango tree's height without climbing it.
Lesson 3 Cosine of an Acute Angle: The Adjacent Side Speaks	Learners observed that the cosine of an acute angle is a fixed ratio (adjacent divided by hypotenuse) that can be read from mathematical tables, that cosine values	The cosine of an acute angle theta equals the length of the adjacent side divided by the length of the hypotenuse of any right-angled triangle containing theta. Like sine,	Cosine, like sine, is a consistent fingerprint of an angle: it compresses the shape of any right-angled triangle with that angle into a single number. Because every right-angled

	decrease as the angle increases, and that for each angle in their comparison table the cosine value appeared to match the sine value of a different angle — specifically the angle that adds up to 90 degrees.	this value depends only on the angle and not on the size of the triangle. For the mango-tree scenario, $\cos 53^\circ$ is approximately 0.6018, meaning the 9.0 m shadow is approximately 0.6018 times the length of the sun-ray reaching the tree tip.	triangle with a 53-degree angle has the same ratio of adjacent to hypotenuse regardless of size, a table entry is sufficient to solve for any unknown side in any such triangle anywhere in Kenya — from a hospital ramp in Nakuru to the shadow of the mango tree in Kisumu.
Lesson 4 Tangent and the Sine–Cosine–Tangent Relationship: Solving the Mango-Tree Problem	Learners observed the algebraic derivation showing that dividing $\sin \theta$ by $\cos \theta$ (each defined using the hypotenuse) causes the hypotenuse to cancel, yielding Opposite/Adjacent — the tangent ratio. They also observed numerical agreement between $\sin \theta / \cos \theta$ and $\tan \theta$ (from tables) for $\theta = 30^\circ, 40^\circ, 60^\circ$, and 70° , confirming the identity across four distinct angles.	Learners learned that $\tan \theta = \text{Opposite} / \text{Adjacent}$ and that this ratio is algebraically equivalent to $\sin \theta / \cos \theta$, making all three trigonometric ratios interdependent. They learned that knowing an acute angle and the length of the adjacent side (e.g., a shadow) is sufficient to calculate the opposite side (e.g., a tree's height) using $h = \text{adjacent} \times \tan \theta$.	Learners explained why the mango tree in the Kisumu school compound is approximately 11.94 m tall: because the sun angle of 53° and the 9.0 m shadow form a right triangle in which $\tan 53^\circ = \text{tree height} / \text{shadow length}$, giving $h = 9.0 \times 1.3270 \approx 11.94$ m. This result is consistent with the similar-triangles cross-check ($h \approx 12.0$ m), confirming that the tangent ratio is a reliable, angle-based tool for indirect measurement.
Lesson 5 Trigonometric Ratios Using Calculators: Precision, Inverse Functions, and the Shadow Puzzle	Learners used scientific calculators to evaluate $\sin$, $\cos$, and $\tan$ for acute angles including 53° (the Kisumu sun angle), and used inverse functions ($\sin^{-1}$, $\cos^{-1}$, $\tan^{-1}$) to recover angles from given ratios. They also computed that $\tan 53^\circ \times 9.0 \text{ m} \approx 11.94$ m — a calculator-precision estimate of the mango tree height.	Learners discovered that: (1) the calculator returns consistent, high-precision trigonometric ratio values that match but are more precise than hand-drawn triangle estimates; (2) inverse trigonometric functions reverse the process — given a ratio, they return the angle; (3) the percentage error between their Lesson 1 hand-drawn estimates and calculator values reflects the physical limitations of rulers and protractors, not a flaw in the mathematics; (4) integrity in mathematics means reporting errors honestly rather than adjusting data.	The calculator does not generate new mathematics — it applies the same consistent ratio relationships between angles and side lengths in right-angled triangles that learners established geometrically in Lessons 1–4. The precision of the calculator-derived values reflects the mathematical exactness of the trigonometric function definition. When applied to the Kisumu phenomenon, $\tan 53^\circ = 1.3270$ allows us to calculate the mango tree height as approximately 11.94 m — without climbing it — advancing the driving question answer with quantitative confidence.
Lesson 6 Complementary Angles and the $\sin \theta = \cos(90^\circ - \theta)$ Identity	Learners read an article about a Nairobi structural engineer using complementary angles, watched a 4-minute offline video showing how Opposite and Adjacent sides swap when a right-angled triangle is viewed from the complementary angle, and completed a five-row table comparing $\sin \theta$ with $\cos(90^\circ - \theta)$ for pairs including the familiar $53^\circ/37^\circ$ Kisumu mango-tree angles.	Learners discovered and proved that $\sin \theta = \cos(90^\circ - \theta)$ for any acute angle θ , because in a right-angled triangle the two acute angles are complementary and the side that is Opposite one angle is Adjacent to the other. They applied this identity to simplify expressions, solve equations, and confirm the Kisumu tree calculation.	The complementary-angle identity means that a surveyor (or a student in Kisumu) only needs one clinometer reading — the sun's elevation angle θ — to calculate the tree height, because $\sin \theta$ and $\cos(90^\circ - \theta)$ reference the same ratio of sides in the same triangle. The identity halves the number of independent trigonometric relationships that need to be memorised, revealing an elegant symmetry at the heart of right-triangle geometry.

Lesson 7 Special Angles (Derivation): Unlocking Exact Trigonometric Values	Learners observed — through their own geometric constructions — that halving an equilateral triangle of side 2 produces a right-angled triangle with sides 1, $\sqrt{3}$, and 2, and that an isosceles right-angled triangle with legs 1 has a hypotenuse of $\sqrt{2}$. They observed that the trigonometric ratios read from these triangles are exact expressions involving $\sqrt{2}$ and $\sqrt{3}$, not messy decimals, and that these values appeared consistently in the video and reference reading.	Learners learned that five special angles (0°, 30°, 45°, 60°, 90°) have exact trigonometric ratios derivable from geometry alone — no calculator needed. They learned that $\sin 30^\circ = 1/2$, $\cos 30^\circ = \sqrt{3}/2$, $\tan 30^\circ = 1/\sqrt{3}$; $\sin 45^\circ = \cos 45^\circ = 1/\sqrt{2}$; $\sin 60^\circ = \sqrt{3}/2$, $\cos 60^\circ = 1/2$, $\tan 60^\circ = \sqrt{3}$; and the boundary values at 0° and 90°. They also learned that $\sin \theta = \cos(90^\circ - \theta)$ for complementary angles, and that $\tan 90^\circ$ is undefined because the Adjacent side of the triangle collapses to zero.	Learners explained how the exact values connect back to the anchoring phenomenon: if the sun in Kisumu made an angle of exactly 60° with the ground, the height of the mango tree (shadow = 9.0 m) would be exactly $9\sqrt{3}$ metres — a precise, unrounded answer. They explained this on their passport cards and in their cumulative models, articulating that exact values allow more reliable calculations in real-world measurement tasks such as surveying, construction, and navigation across Kenya.
Lesson 8 Special Angles: Solving Real Problems Without a Calculator	Learners folded equilateral and isosceles right-triangle card cutouts, measured and labelled all sides using Pythagoras, and read off exact sin, cos, and tan values for 30°, 45°, and 60°. They verified these values against calculator outputs and self-marked their Phase 1 prediction tables.	For special angles (0°, 30°, 45°, 60°, 90°), the trigonometric ratios are exact values derivable from the geometry of equilateral and isosceles right triangles — no calculator is required. The identity $\tan \theta = \sin \theta / \cos \theta$ holds for every special angle. These exact values can be applied directly to real problems such as calculating roof-rafter lengths for a 60°-pitched roof in Eldoret.	Using exact special-angle values and the relationship $\tan \theta = \sin \theta / \cos \theta$, learners can calculate unknown side lengths in any right-angled triangle whose acute angle is a special angle — including the height of the Kisumu mango tree if the sunbeam angle were exactly 60°. This advances the driving question by showing that the angle–ratio relationship is not just consistent and calculator-dependent, but for special angles it is perfectly exact and derivable from pure geometry.
Lesson 9 Angles of Elevation and Depression: Reading Heights from the Sky	We observed that when a student looks up at the top of the school flagpole and measures the angle with a clinometer, the horizontal distance to the base and that angle are enough information to calculate the height — without climbing the pole or using a shadow.	We learned that the angle of elevation is measured upward from the horizontal to the line of sight toward a higher object, the angle of depression is measured downward from the horizontal to the line of sight toward a lower object, and in both cases the tangent ratio connects the angle to the opposite side (height or depth) and the adjacent side (horizontal distance).	We explained these observations using the tangent ratio: height equals baseline times $\tan(\text{angle of elevation})$. This is the same relationship used in Lesson 4 to find the mango-tree height from the sun angle, confirming that the driving question can be answered for any tall object — tree, flagpole, water tank, or aircraft on a glide path — as long as we can measure an angle and a baseline distance.
Lesson 10 Real-Life Applications, DQB Closure & Final Model Revision	Students observed that the same trigonometric ratios that describe Achieng's shadow-and-tree relationship in Kisumu also govern the angle of a plot boundary fence in Kisumu (Station A), the height of a Safaricom telecommunications mast in Kericho calculated exactly using $\tan 60^\circ = \sqrt{3}$ (Station B), and the glide-path distance of an aircraft approaching Wilson Airport where the complementary-angle identity $\sin$	Students learned that trigonometric ratios (sin, cos, tan and their reciprocals cosec, sec, cot) form a unified, interconnected system — not isolated formulas — in which SOH CAH TOA, special-angle exact values, and the complementary identity $\sin \theta = \cos(90^\circ - \theta)$ are all expressions of the same underlying geometric relationship between angles and side-length ratios in right-angled triangles. They learned that this relationship	Students explained that the angle a sunbeam makes with the ground (the angle of elevation) creates a right-angled triangle whose sides are related by fixed trigonometric ratios regardless of the triangle's size, because all triangles with the same angle are similar and therefore preserve the ratio of any two corresponding sides. Therefore, once you know the angle and one side length (a shadow, a measured

	$3^\circ = \cos 87^\circ$ was verified numerically (Station C). They also observed, by holding their Lesson 1 and Lesson 10 models side by side, the measurable growth in their own understanding across the unit.	is scale-invariant: it works identically whether the triangle is the small one formed by Achieng's shadow or the large one formed by an aircraft's glide path, because similar triangles preserve ratios. They confirmed that the mango tree's height (≈ 12.0 m) can be found by at least two independent trigonometric methods, validating the consistency of the ratio system.	distance, or an altitude), the trigonometric ratio — sin, cos, or tan — acts as a conversion factor that scales any known length into the unknown height or distance. This is why Achieng, a land surveyor in Kisumu, a Safaricom engineer in Kericho, and a pilot approaching Wilson Airport all use the same mathematical tool: the consistent, angle-determined ratios encoded in SOH CAH TOA.
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Production & Layout: Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

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